

Better computer use. Measure the whole loop.



Slack workflows



Independent evaluation

- Goal: complete useful work with fewer errors, less time and lower cost.
- 18 tasks covering search, threads, edits, DMs, pins, saved messages and channel details.
- Relay combines a runnable Slack-like app, a bounded harness and independent checks.
- Improve the model, interface and harness separately. Keep the evidence for each change.



Task + seed



Private workspace



Agent actions



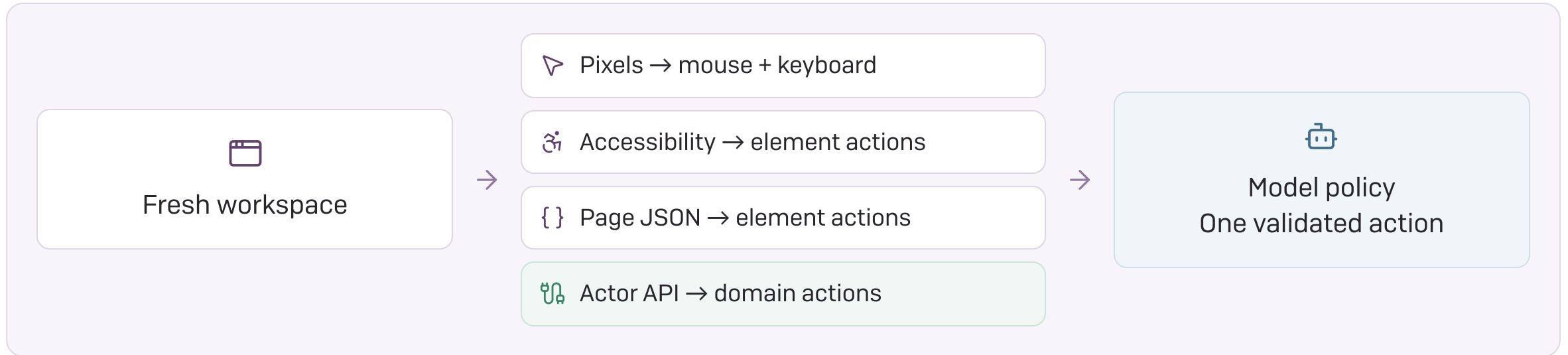
State evaluation

The harness connects action to evidence

- Observe the workspace. Ask for one action. Validate it before execution.
- Execute with Playwright or the actor API. Record the request, response and state change.
- Repeat within time, action and spend limits. Grade the final state and export the trace.
- React renders the workflows. A shared transition engine writes private SQLite state.

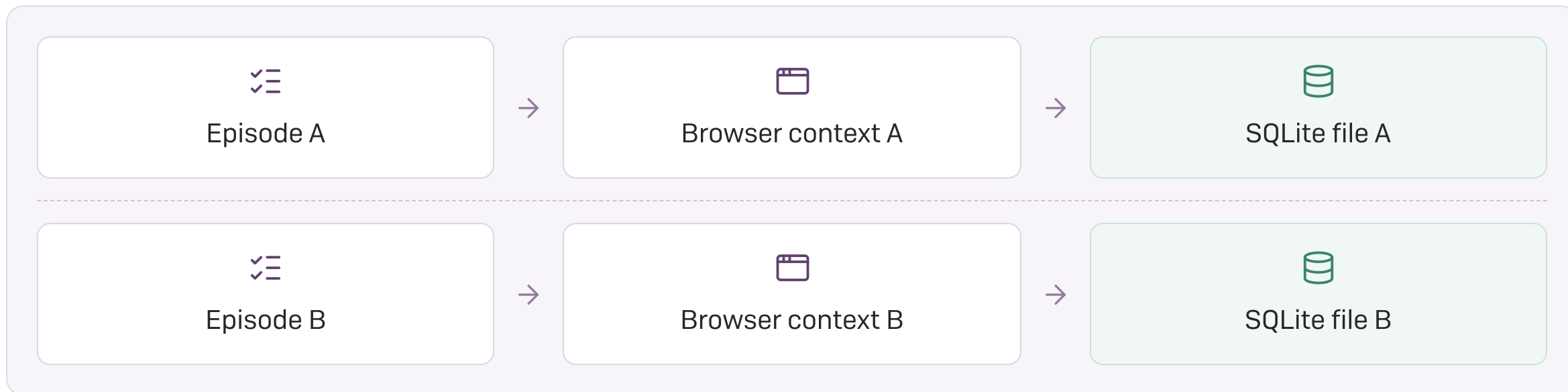
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- MIT for original code; third-party asset restrictions remain separate.
 - Cua Slack, env0 and Agent-Diff address related environments. This is not a novelty claim.
 - Source review, not comparative execution: [prior-work survey and pinned references](#).

Four observation and action interfaces



- The actor never receives grader answers or control credentials.
- API access changes information and action granularity; it is not a computer-use score.
- Optional documentation and bounded history are explicit conditions, not hidden prompt changes.

Episode lifecycle and session isolation



- Fresh contexts and private files prevent cross-episode state contamination.
- Transactional writes, revisions and request IDs enforce atomicity and idempotency.
- Actor and control listeners share a process: this is data isolation, not a hostile-code sandbox.

Set up once. Change the model route.

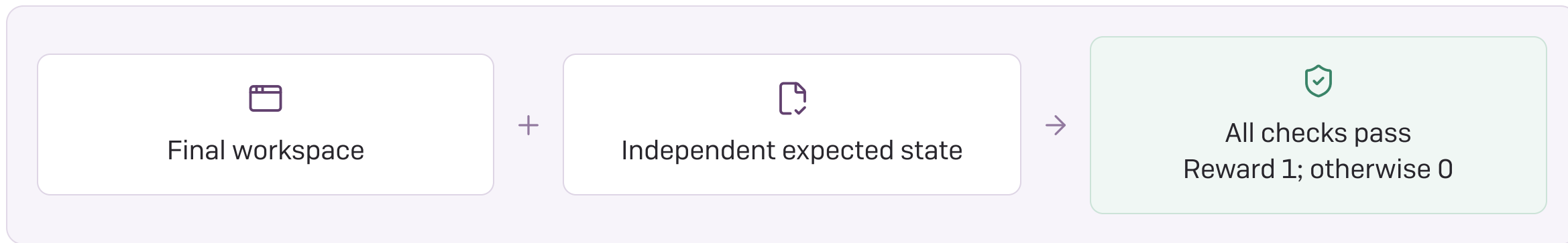
 17 model routes 18 tasks 1 attempt per cell

- Node 24: install dependencies and Chromium, build, then run Relay Live. Exact commands are in the README.
- Connect a Router key and select an available model. The same open-source Playwright harness serves all 17 routes.
- Fresh workspaces; seed 1042; accessibility control; recent-four history; no guide.
- 40 actions; 4,096 output tokens per call; 180 seconds and \$5 maximum per trial.
- 105 prior attempts retained; only 201 untouched cells continued. No repeats or dropped failures.
- \$300 shared cap includes prior pilots. No hidden retry or provider substitution.
- Public tasks and temporal cohorts: coverage evidence, not a reliable model ranking.

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- [Frozen plan, preserved origins and explicit scheduling amendment.](#)

Turn a workflow into a benchmark contract

- Six control tasks; 12 deeper workflows with 3–6 coordinated mutations.
- Decision record: retrieve final approval, update description, pin the root, save the reply and acknowledge in-thread.
- Check content, identity, author, destination, uniqueness and all unchanged workspace state.
- Wrong thread, replacement instead of edit, stale fact or collateral change fails.



- Exact text is deliberate. Fully undone actions may pass; path-level safety is not currently graded.

Verification separates software checks from model runs

- 174 backend/harness checks and 91 browser checks.
- 84 valid states and 2,583 adversarial invalid states for the 12 deep-task graders.
- Scripted UI recipes exercise interaction paths; they do not measure model capability.
- 306 / 306 recorded model/task attempts in the one-pass comparison; earlier interface trials remain separate.



Software tests
Does the app work?



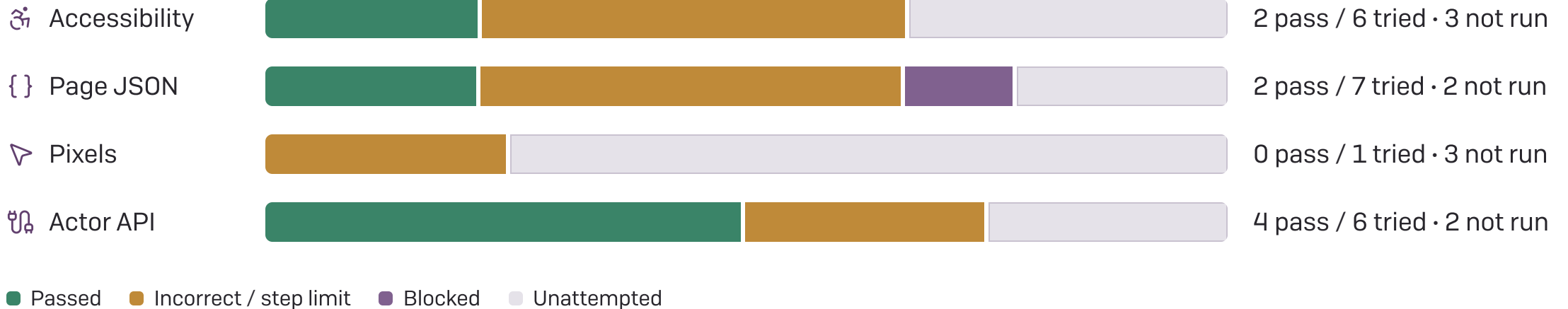
Grader challenges
Does reward mean success?



Model episodes
Can the policy do the task?

Earlier interface study · separate from the 306 cells

- 20 attempted: 8 passed, 11 incorrect/step-limited, 1 blocked; 10 unattempted.
- 126 requests; \$0.149874 estimated including \$0.001490 reserved for the blocked response.



- Unequal task/model coverage; these totals are not causal interface comparisons. Cost is not an invoice.
- [Complete inventory, exact trajectories and hashes.](#)

17 models. Every task.

1 per task/model · Collection complete

306 / 306 attempted · 152 passed · 61 incomplete · 93 blocked

\$189.3110

Usage estimate

\$3.7794

Unresolved · 35 calls

\$193.0904

Total allowance · 306 trials

Model ↕	Trials ↕	Passed ↕	Incomplete ↕	Blocked ↕	Median time ↕	Usage estimate ↕	Unresolved ↕	Total allowance ↕
 GPT-6 Astra	18 / 18	14 / 18	1	3	27.1s	\$35.9623	\$0.0000	\$35.9623
 GPT-6.1 Sol	18 / 18	14 / 18	4	0	28.5s	\$8.4352	\$0.0000	\$8.4352
 GPT-6 Luna	18 / 18	9 / 18	8	1	39.4s	\$0.5394	\$0.0092	\$0.5486
 GPT-OSS 120B	18 / 18	5 / 18	2	11	180.1s	\$0.6682	\$0.1358	\$0.8040
 Claude Fable 5.1	18 / 18	13 / 18	1	4	62.8s	\$48.4714	\$0.0000	\$48.4714
 Claude Opus 5.5	18 / 18	13 / 18	4	1	38.3s	\$28.6987	\$0.3741	\$29.0728
 Claude Sonnet 5.5	18 / 18	7 / 18	4	7	39.9s	\$13.9323	\$0.3304	\$14.2627
 Claude Haiku 4.5	18 / 18	6 / 18	3	9	28.3s	\$3.7921	\$0.0000	\$3.7921
 Grok 4.7	18 / 18	10 / 18	6	2	34.3s	\$11.9101	\$0.3809	\$12.2910
 Qwen 3.8 Max	18 / 18	7 / 18	1	10	125.1s	\$6.9999	\$1.0639	\$8.0638
 DeepSeek V4 Pro	18 / 18	8 / 18	1	9	106.8s	\$4.6816	\$0.2547	\$4.9363
 DeepSeek V4.1 Flash	18 / 18	7 / 18	3	8	65.9s	\$1.3825	\$0.0541	\$1.4365
 GLM 5.3	18 / 18	7 / 18	0	11	125.3s	\$3.6307	\$0.3453	\$3.9760
 GLM 5.3 Flash	18 / 18	8 / 18	0	10	46.2s	\$0.4444	\$0.0000	\$0.4444
 Kimi K3	18 / 18	11 / 18	3	4	50.0s	\$14.0051	\$0.7958	\$14.8009
 MiniMax M3	18 / 18	5 / 18	10	3	25.9s	\$1.4698	\$0.0353	\$1.5052
 Nemotron 3 Ultra	18 / 18	8 / 18	10	0	50.1s	\$4.2874	\$0.0000	\$4.2874

Passed Incomplete Blocked Not run

- 18 public tasks · 1 attempt per task/model · accessibility control · \$300 shared ceiling. Median time includes blocks. Snapshot 2026-10-03 21:14 UTC.
- Accepted receipts: 74,861,087 input + 1,400,260 output tokens; 4,712 receipts / 4,747 calls. Base-rate estimates, not invoices; missing tokens are unknown. Shared ledger: \$194.1058 including \$1.0154 outside these 306 cells; \$105.8942 remains under \$300. [Full costs CSV](#) · [JSON](#)
- 105 prior attempts preserved, including failures. New cells only; no retries. Family/tier coverage, not a popularity ranking. Raw outcomes; release-sync and design-handoff have wording caveats.
- [Frozen design](#) · [Collection status and evidence](#) · [Download all trials \(CSV\)](#) · [Earlier 36-trial pilot: 20 passed](#)

Why did runs pass, fail or stop?

- Thread repair: 13 / 17 passed; handoff DM: 16 / 17. Thread repair combines fact retrieval, an original-message edit, a reaction and a reply.
- Saved-item cleanup: 4 / 17 strict passes. A further diagnostic workspace pass ended at an output limit and remains blocked.
- Design handoff: 0 / 17. Six completed runs copied the quoted “DESIGN”; the grader expected the retrieved name. The substitution instruction was implicit.
- Release-sync: 0 / 17, with an ambiguous source-location hint—not clean capability evidence. Sol made partial changes, then repeated edit/cancel actions.

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- [Trace identities and outcome details](#). One attempt per public task; no claim of repeatability or a causal model ranking.

Speed and resource measurements



17.34 ms

Create session · p50



1.50 ms

Reset state · p50



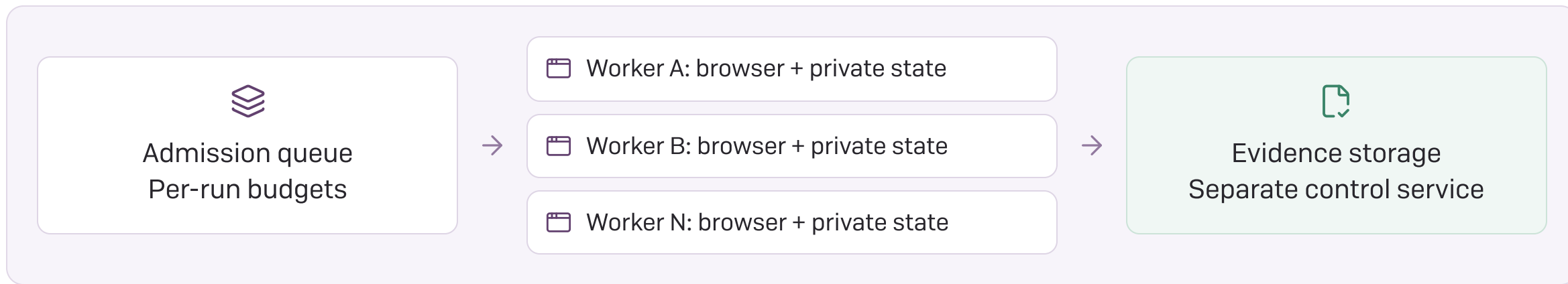
84.15 ms

Screenshot · p50

- 100 persisted sessions, 400 reads at concurrency 16, and 10 sequential browser contexts.
- Machine: Apple M5 Max, Node 24; timings include the stated operation only.
- Browser child-process memory, sustained agent concurrency and multi-host capacity are not measured.
- Model calls and visual evidence add latency; API throughput is not agent throughput.

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- [Raw samples, machine details and measurement boundaries.](#)

What I learned · what to build next



- A valid click is not task success. Check intended changes and the state that must stay unchanged.
- Task wording is part of the test. Version ambiguous instructions before collecting new results.
- Missing receipts, blocked runs and wrong answers mean different things. Preserve all three.
- Scale with bounded workers and external evidence storage. Measure full worker memory, p95 latency and cleanup first; this diagram is a proposal.

Demo, then questions

- [Open the workspace](#): search, reply in a thread, and edit your own message.
 - [Open trial review](#): choose a recorded model/task, play a success, and inspect an incomplete or blocked attempt.
 - Inspect the exact request, action, state change and terminal checks; replay requires no model calls.
 - Which Slack workflows and visual details matter for the target agent?
 - What should reward penalize beyond final state? What isolation, throughput and cost target do we need?
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- Reset/step/reward are implemented; no RL training or BrowserGym/OpenEnv compatibility is claimed.
 - Next gates: successful pixel policies, held-out task templates, trainer integration and sustained worker tests.
 - [Code and setup](#) · [Submission assessment and limitations](#).